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Time taken	10 days 20 hours
Marks	30.00/30.00
Grade	**100.00** out of 100.00
Name	DHIRU BHAI 730423104017

Question 1

Correct

Mark 10.00 out of 10.00

Create an ArrayList which will be able to store only Strings. Create a printAll method which will print all the elements using an Iterator.

Note: Size of ArrayList is not fixed. Continue till you receive the word END

Sample Input:

Java Oracle C MySql Perl Pascal END

Sample Output:

Java Oracle C MySql Perl Pascal

For example:

Input	Result
Java Oracle C MySql Perl Pascal END	Java Oracle C MySql Perl Pascal

Answer: (penalty regime: 0 %)

```
import java.util.ArrayList;
import java.util.Iterator;
import java.util.Scanner;

public class StringList {
    private ArrayList<String> stringList;

    public StringList() {
        stringList = new ArrayList<>();
    }

    public void addString(String str) {
        stringList.add(str);
    }

    public void printAll() {
        Iterator<String> iterator = stringList.iterator();
        while (iterator.hasNext()) {
            System.out.print(iterator.next() + " ");
        }
        System.out.println(); // for a new line at the end
    }

    public static void main(String[] args) {
        StringList list = new StringList();
        Scanner scanner = new Scanner(System.in);

        while (true) {
            String input = scanner.nextLine();
            if (input.equals("END")) {
                break;
            }
            list.addString(input);
        }

        list.printAll();
        scanner.close();
```

```
38      }
39  }
```

	Input	Expected	Got	
✔	Java Oracle C MySql Perl Pascal END	Java Oracle C MySql Perl Pascal	Java Oracle C MySql Perl Pascal	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 2

Correct

Mark 10.00 out of 10.00

Develop a java class Collections_ArrayList and perform the following operations.

1. Create a method saveEvenNumbers(int N) and use ArrayList to store even numbers from 2 to N, where N is a integer which is passed as a parameter to the method saveEvenNumbers(). The method should return the ArrayList (A1) created.
2. In the same class create a method printEvenNumbers() which iterates through the arrayList A1 in step 1, and It should multiply each number with 2 and display it in format 4,8,12....2*N.

Note: Create ArrayList A1 and A2 as instance variables.

For example:

Input	Result
23	List of Even Numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22] List of Even Numbers multiplied by 2: [4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44]
78	List of Even Numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78] List of Even Numbers multiplied by 2: [4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68, 72, 76, 80, 84, 88, 92, 96, 100, 104, 108, 112, 116, 120, 124, 128, 132, 136, 140, 144, 148, 152, 156]

Answer: (penalty regime: 0 %)

```
import java.util.ArrayList;
import java.util.Scanner;
public class Collections_ArrayList
{
    private ArrayList<Integer> A1;
    private ArrayList<Integer> A2;
    public Collections_ArrayList()
    {
        A1 = new ArrayList<>();
        A2 = new ArrayList<>();
    }
    public ArrayList<Integer> saveEvenNumbers(int N)
    {
        A1.clear();
        for (int i = 2; i <= N; i += 2)
        {
            A1.add(i);
        }
        return A1;
    }
    public ArrayList<Integer> printEvenNumbers()
    {
        A2.clear();
        for (int num : A1)
        {
            A2.add(num * 2);
        }
    System.out.println("List of Even Numbers:\n" + A1);
System.out.println("List of Even Numbers multiplied by 2:\n"+ A2);
    return A2;
    }
    public static void main(String[] args)
```

```
    {
        Collections_ArrayList obj = new Collections_ArrayList();
        Scanner scanner = new Scanner(System.in);
        int N = scanner.nextInt();
        obj.saveEvenNumbers(N);
        obj.printEvenNumbers();
    }
}
```

	Input	Expected	Got	
✔	23	List of Even Numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22] List of Even Numbers multiplied by 2: [4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44]	List of Even Numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22] List of Even Numbers multiplied by 2: [4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44]	✔
✔	78	List of Even Numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78] List of Even Numbers multiplied by 2: [4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68, 72, 76, 80, 84, 88, 92, 96, 100, 104, 108, 112, 116, 120, 124, 128, 132, 136, 140, 144, 148, 152, 156]	List of Even Numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78] List of Even Numbers multiplied by 2: [4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68, 72, 76, 80, 84, 88, 92, 96, 100, 104, 108, 112, 116, 120, 124, 128, 132, 136, 140, 144, 148, 152, 156]	✔

Passed all tests! ✔

▸ **Show/hide question author's solution (Java)**

Correct

Marks for this submission: 10.00/10.00.

Question 3

Correct

Mark 10.00 out of 10.00

Write a program to read a set of strings during runtime using a Hashset S1, LinkedHashSet S2 and TreeSet S3 which will be able to store only Strings. Create a printAll() method and pass S1, S2, S3 which will print all the elements and observe the inference.

Note: Size of the set is 6.

For example:

Input	Result
Infosys HCL Wipro TechMahindra Hexaware Altran	Using HashSet: [HCL, Wipro, Infosys, TechMahindra, Altran, Hexaware] Using LinkedHashSet: [Infosys, HCL, Wipro, TechMahindra, Hexaware, Altran] Using TreeSet: [Altran, HCL, Hexaware, Infosys, TechMahindra, Wipro]
AAA BBB CCC DDD EEE FFF	Using HashSet: [AAA, CCC, BBB, EEE, DDD, FFF] Using LinkedHashSet: [AAA, BBB, CCC, DDD, EEE, FFF] Using TreeSet: [AAA, BBB, CCC, DDD, EEE, FFF]

Answer: (penalty regime: 0 %)

```
import java.util.HashSet;
import java.util.LinkedHashSet;
import java.util.Scanner;
import java.util.Set;
import java.util.TreeSet;

public class SetExample {

    // Method to print all elements of the sets
    public static void printAll(Set<String> s1, Set<String> s2, Set<String>
        System.out.println("Using HashSet: " + s1);
        System.out.println("Using LinkedHashSet: " + s2);
        System.out.println("Using TreeSet: " + s3);
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Read input

        String input = scanner.nextLine();
        String[] elements = input.split(" ");

        // Initialize sets
        HashSet<String> s1 = new HashSet<>();
        LinkedHashSet<String> s2 = new LinkedHashSet<>();
        TreeSet<String> s3 = new TreeSet<>();

        // Add elements to set
        for (String element : elements) {
            if (s1.size() < 6) {
                s1.add(element);
                s2.add(element);
                s3.add(element);
            }
        }


```

```
39          printAll(s1, s2, s3);
40
41          scanner.close();
42      }
43  }
```

	Input	Expected	Got	
✔	Infosys HCL Wipro TechMahindra Hexaware Altran	Using HashSet: [HCL, Wipro, Infosys, TechMahindra, Altran, Hexaware] Using LinkedHashSet: [Infosys, HCL, Wipro, TechMahindra, Hexaware, Altran] Using TreeSet: [Altran, HCL, Hexaware, Infosys, TechMahindra, Wipro]	Using HashSet: [HCL, Wipro, Infosys, TechMahindra, Altran, Hexaware] Using LinkedHashSet: [Infosys, HCL, Wipro, TechMahindra, Hexaware, Altran] Using TreeSet: [Altran, HCL, Hexaware, Infosys, TechMahindra, Wipro]	✔
✔	AAA BBB CCC DDD EEE FFF	Using HashSet: [AAA, CCC, BBB, EEE, DDD, FFF] Using LinkedHashSet: [AAA, BBB, CCC, DDD, EEE, FFF] Using TreeSet: [AAA, BBB, CCC, DDD, EEE, FFF]	Using HashSet: [AAA, CCC, BBB, EEE, DDD, FFF] Using LinkedHashSet: [AAA, BBB, CCC, DDD, EEE, FFF] Using TreeSet: [AAA, BBB, CCC, DDD, EEE, FFF]	✔

Passed all tests! ✔

▸ **Show/hide question author's solution (Java)**

Correct

Marks for this submission: 10.00/10.00.